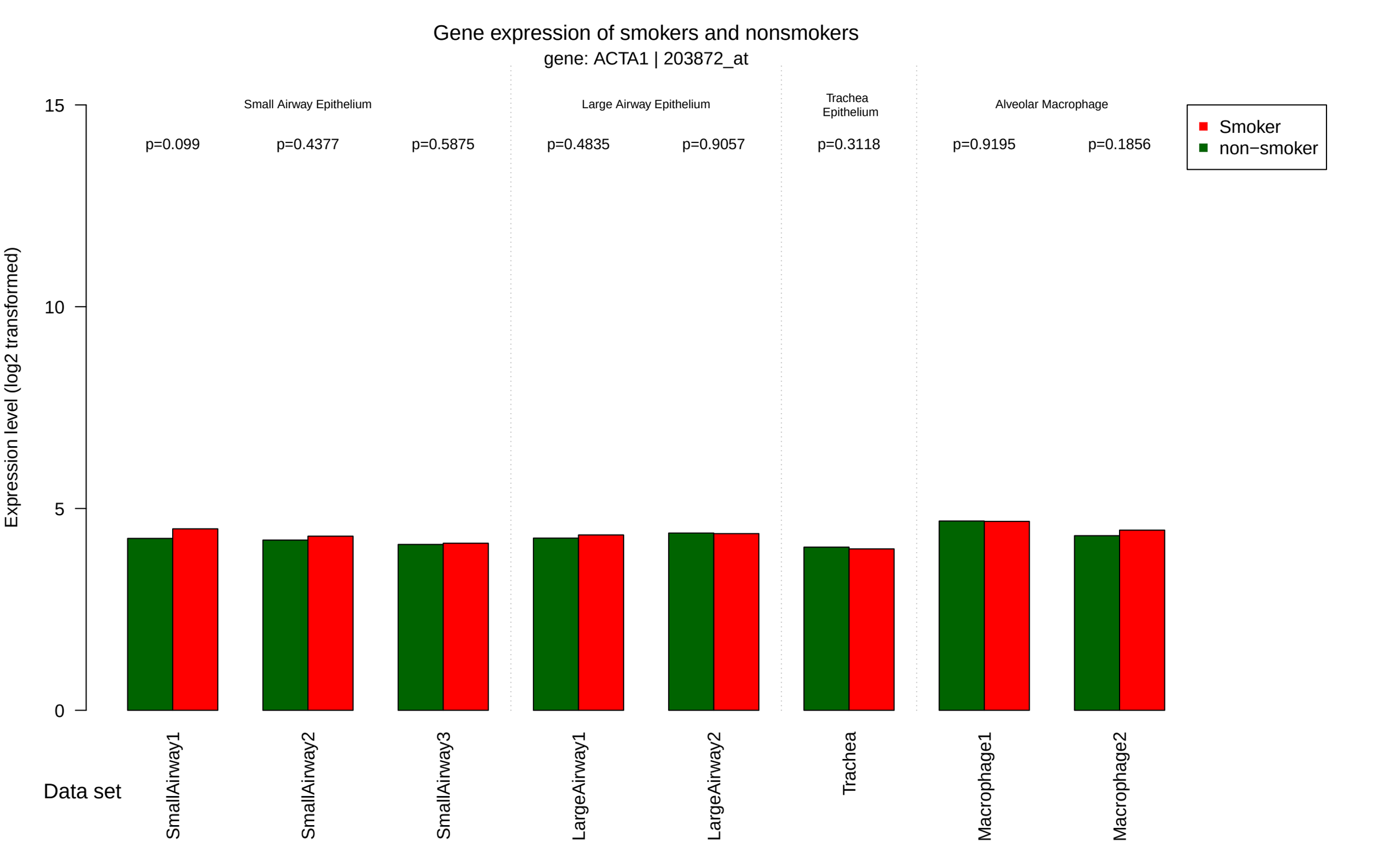
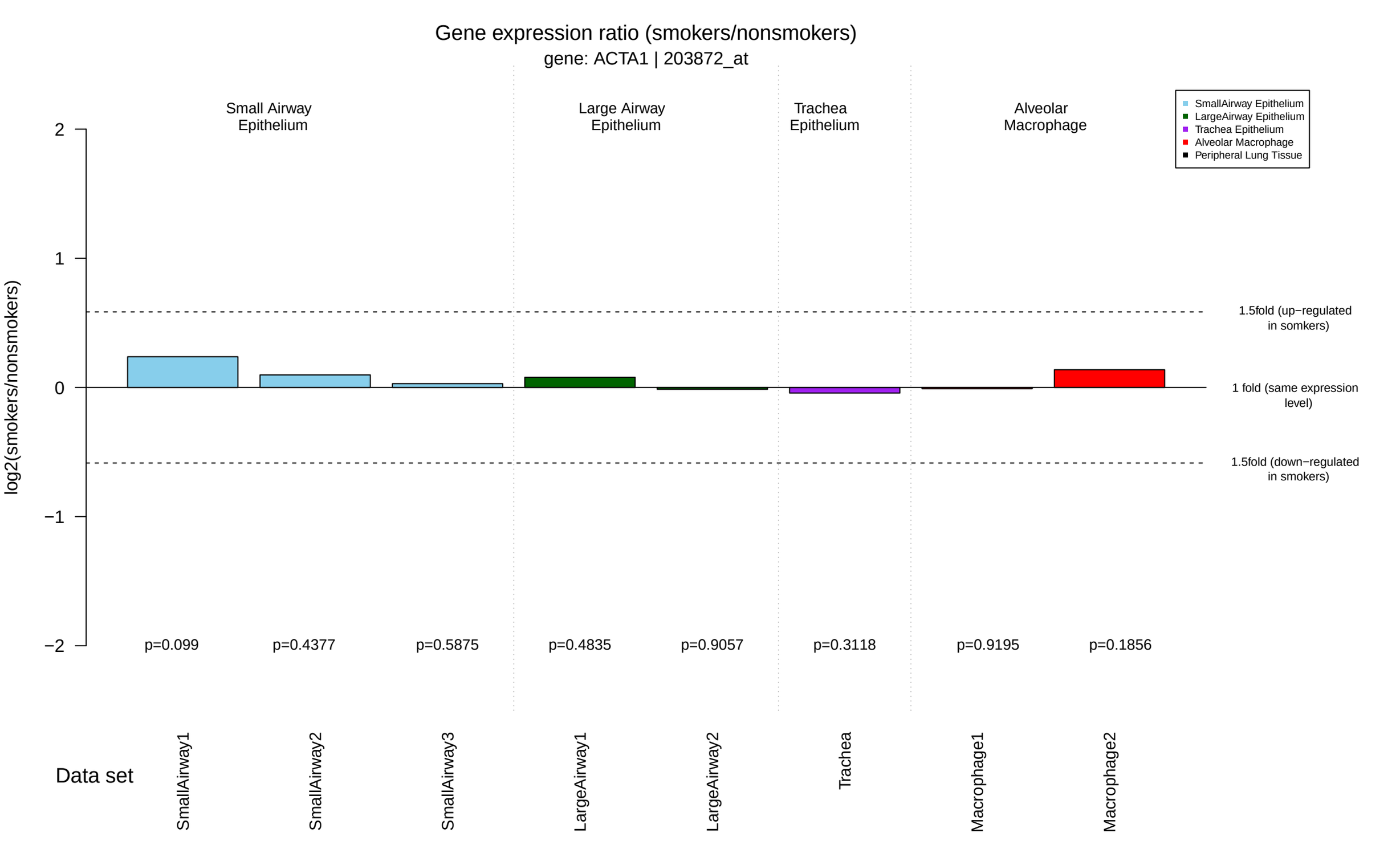
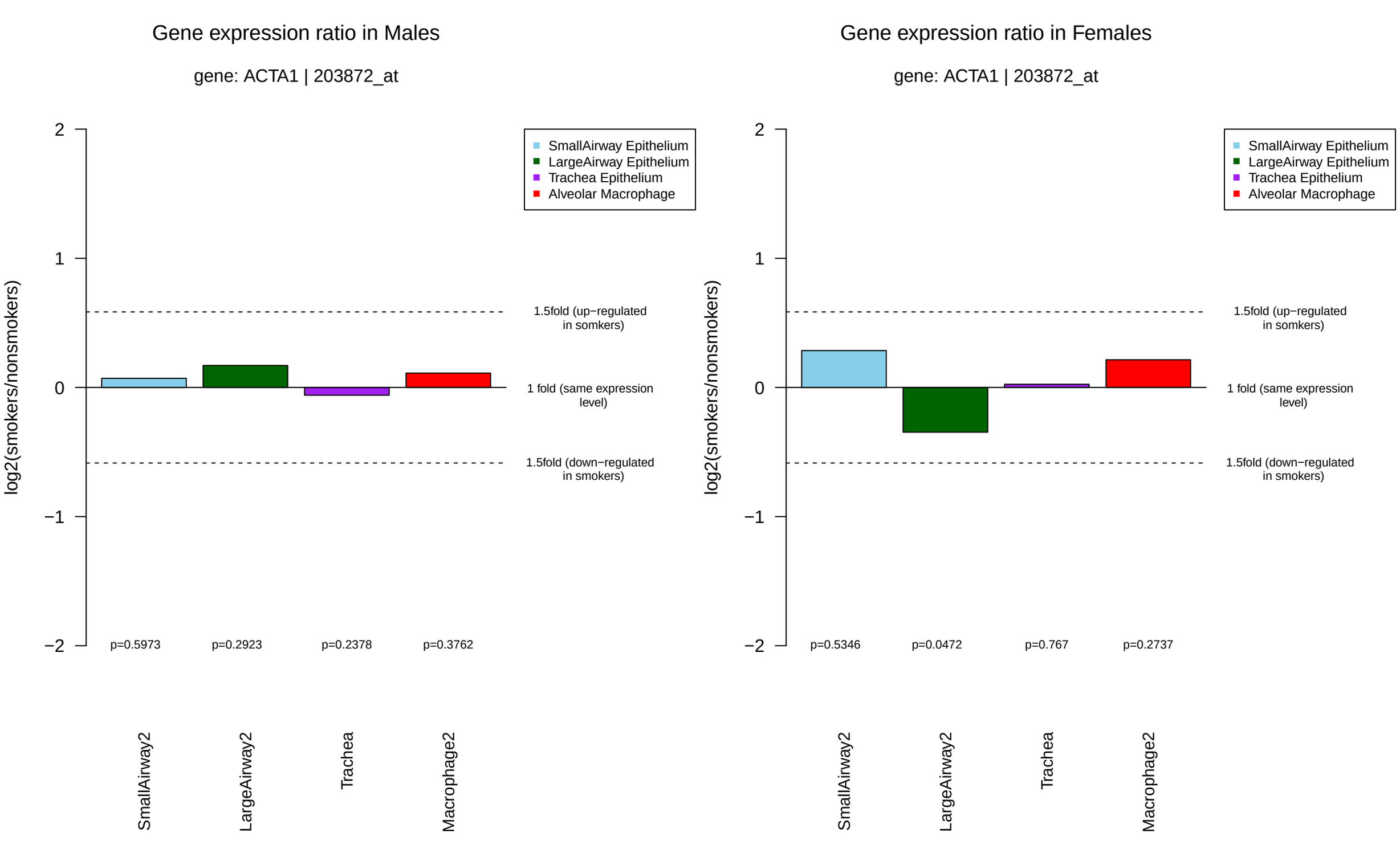
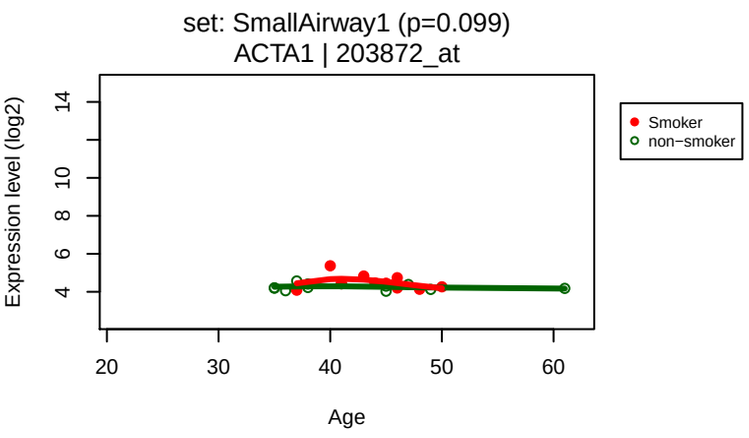
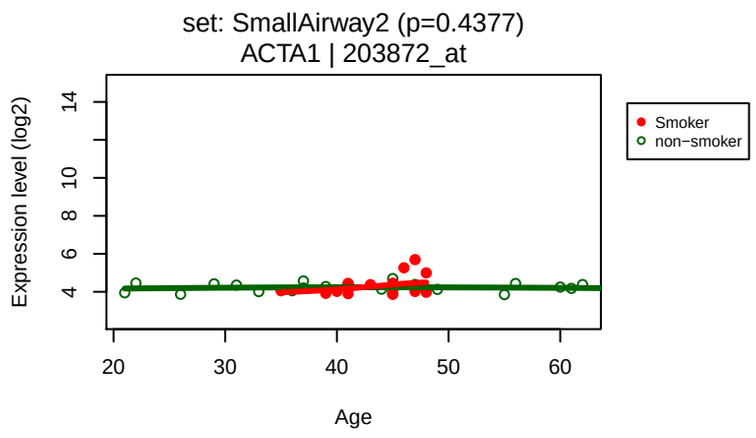
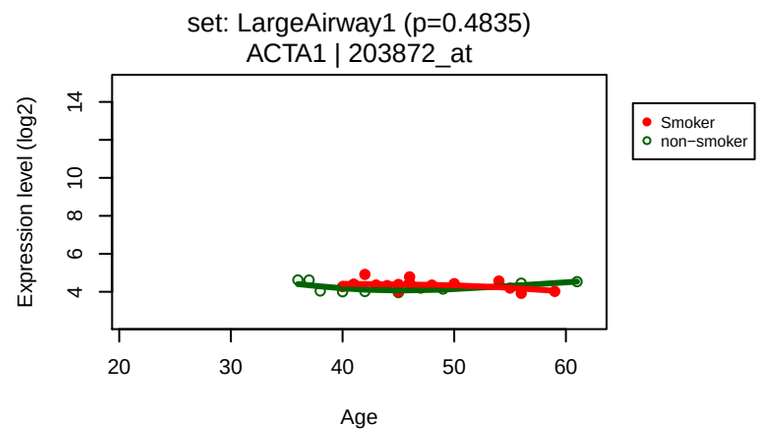

Gene expression of smokers and nonsmokers
gene: ACTA1 | 203872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.099
p=0.4377
p=0.5875
p=0.4835
p=0.9057
p=0.3118
p=0.9195
p=0.1856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

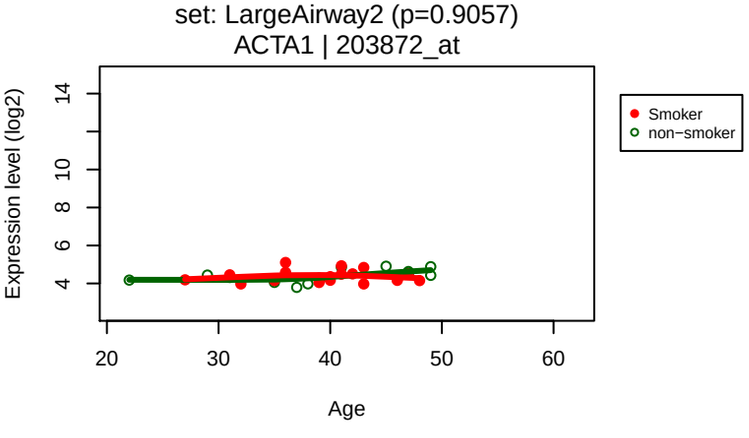

Gene expression ratio (smokers/nonsmokers)
gene: ACTA1 | 203872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.099
p=0.4377
p=0.5875
p=0.4835
p=0.9057
p=0.3118
p=0.9195
p=0.1856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

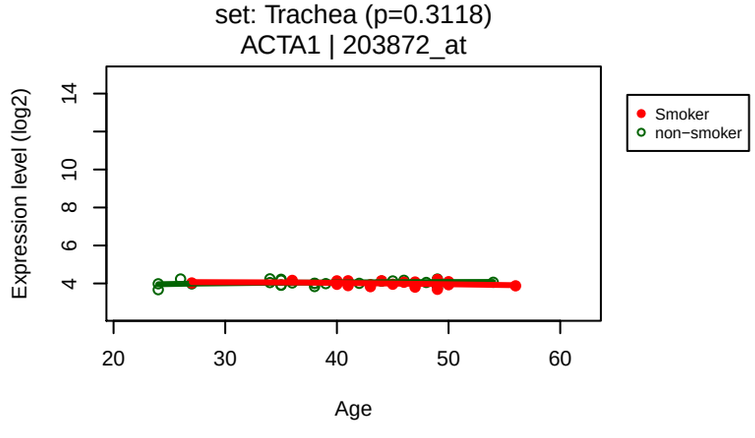

Gene expression ratio in Males
gene: ACTA1 | 203872_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5973
p=0.2923
p=0.2378
p=0.3762
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTA1 | 203872_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5346
p=0.0472
p=0.767
p=0.2737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

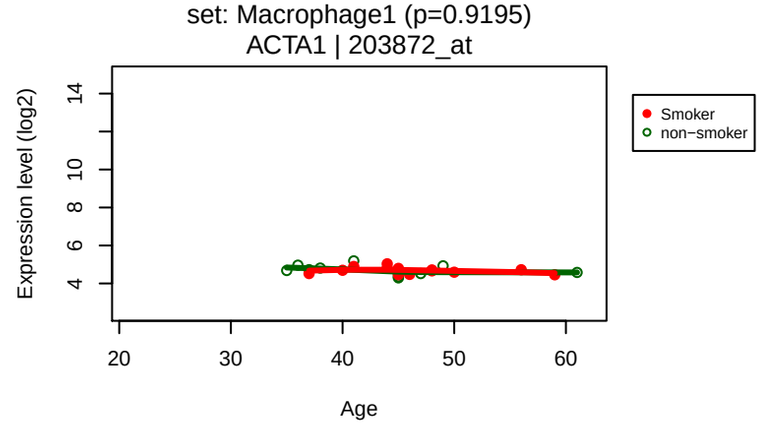

set: SmallAirway1 (p=0.099)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker

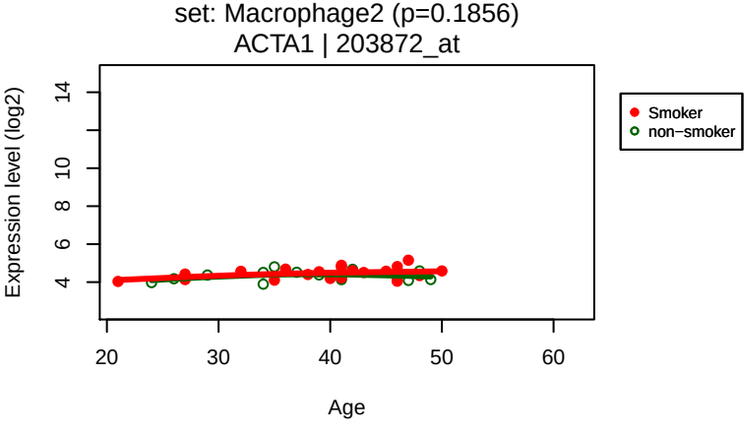

set: SmallAirway2 (p=0.4377)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker

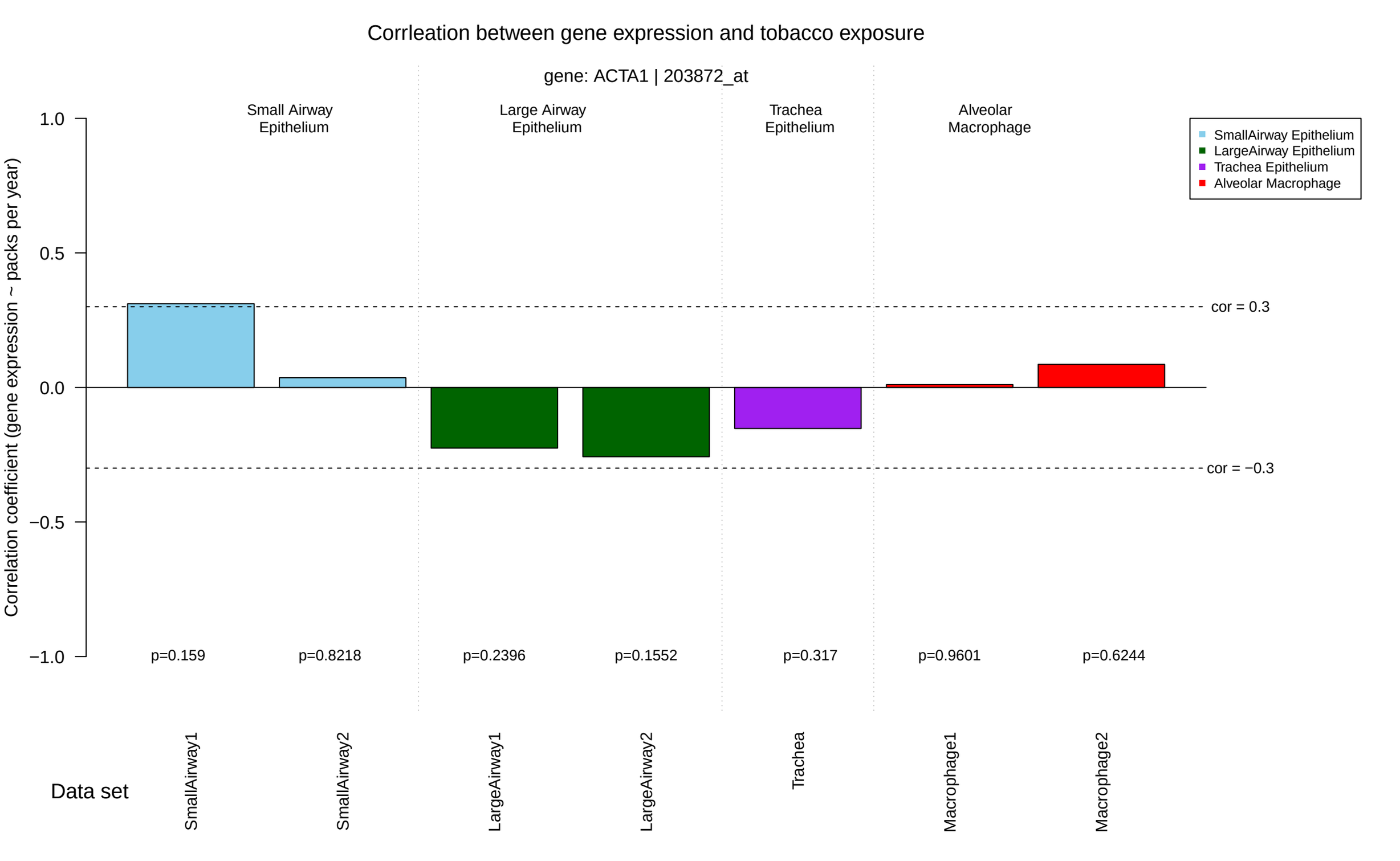

set: LargeAirway1 (p=0.4835)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9057)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3118)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9195)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1856)
ACTA1 | 203872_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTA1 | 203872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.159
p=0.8218
p=0.2396
p=0.1552
p=0.317
p=0.9601
p=0.6244
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2